

BOB-015 – Plugin degenerate-input crash audit

Revision: 1 **Last modified:** 2026-06-09T00:00:00Z **Mandate:** Â§11.4.118 discovery-pressure (confirm known-issue-set completeness; surface unreported crash-on-degenerate-input gaps before end users hit them) **Method:** read-only enumeration (no source/test edits); throwaway probe run from / tmp, then removed.

Methodology

Reproduced the existing offline-isolation harness pattern from tests/unit/test_plugin_crash_guards.py + tests/unit/test_plugin_parser_guards.py: stub novaprinter (capture prettyPrinter rows) and helpers (retrieve_url, download_file, htmlentitydecode, unescape, get_torrent_socks_proxy) into sys.modules, importlib-load each plugin, resolve its class by attribute match (nova2 contract), patch retrieve_url to return each degenerate payload, call search("linux"), and record CRASH vs OK. Run with python3 3.9.x. pytest suite NOT run. No plugins/, tests/, or source files edited.

Probed (15): curated set that ships as files – eztv limetorrents piratebay solidtorrents torlock rutracker rutor kinozal nnmclub – plus broad public plugins anilibra bitsearch gamestorrents megapeer nyaa torrentgalaxy torrentkitty. Excluded the 3 already under active repair this session (tokyotoshokan, kickass, yts). The curated names jackett, torrentproject, torrentscsv do NOT ship as plugin files – out of scope.

Input classes (9): empty_str, whitespace, truncated_html, malformed_html, empty_json {}, empty_json_array [], malformed_json, json_data_null '{"data": null}', none (None).

Result matrix (summary)

Every probeable plugin handled all 8 **string** input classes gracefully (0 rows, no exception) – with ONE exception: piratebay on json_data_null. The none input crashed several plugins but is **not reachable** in production (see reachability note).

- All 14 string-probeable plugins: empty_str / whitespace / truncated_html / malformed_html / {} / [] / malformed_json – **OK** everywhere.
- json_data_null – **OK everywhere EXCEPT piratebay (CRASH).**
- none – CRASH in eztv, limetorrents, piratebay, solidtorrents, torlock, bitsearch, nyaa, torrentgalaxy; OK in rutor, kinozal, nnmclub, anilibra, gamestorrents, megapeer, torrentkitty.
- rutracker – **SKIP-with-reason:** constructor performs a live login (ValueError: Unable to connect using given credentials.), cannot be driven offline. Not a crash.

Reachability note (§11.4.6)

plugins/helpers.py::retrieve_url is typed -> str and every return path returns a str (" " on URLError; dataStr on success) â€” it NEVER returns None. So the none-class crashes are **NOT reachable** via the production helpers path; they are defensive observations, not user-facing defects.

NEW reachable guard gap (beyond tokyotoshokan/kickass/yts)

GAP-1 "plugins/piratebay.py crashes on realistic upstream {"data": null}

Reachability: REAL – retrieve_url can legitimately return a JSON *object* instead of the expected array (same upstream-shape class anilibra was already hardened against in test plugin crash guards.py).

Root cause (piratebay.py search()): json.loads('{ "data": null}')

```
{ "data": None};
```

the guard `if len(response_json) == 0` sees `len == 1` and does NOT fire; for result in response_json: then iterates dict **keys** so `result == "data"` (a str), and `result["info hash"]` indexes a str with a str.

Verbatim captured evidence (retrieve url -> '{"data": null}')

Traceback (most recent call last):

File "<stdin>", line 23, in <module>

File "plugins/piratebay.py", line 91, in search

```
if result["info hash"] == "00000000000000000000000000000000000000000000":
```

```
TypeError: string indices must be integers
```

Suggested fix direction: after `json.loads`, if `not isinstance(response_json, list)`: return before the empty-check / iteration “mirroring the existing anilibra non-list/null-data guard.

Defensive observations (NOT reachable â€” no fix needed)

```

none-only crashers, verbatim messages: eztv / limetorrents / solidtorrents / torlock / nyaa â†’
TypeError: can only concatenate str (not "NoneType") to str;
bitsearch / torrentgalaxy â†’ TypeError: expected string or bytes-like
object; piratebay (none) â†’ TypeError: the JSON object must be str,
bytes or bytearray, not NoneType.

```

Bottom line

Exactly **one** additional reachable gap beyond the three already fixed: `piratebay on {"data": null}`. No other reachable crash surfaced across 135 (plugin $\tilde{\text{A}}$ —input) cells. Honest coverage gaps: `rutracker` (live-login constructor $\hat{+}$ SKIP) and the un-shipped `jackett / torrentproject / torrentscsv` curated names were not exercised.